

# EDUVOS

Faculty of Information Technology

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## RESEARCH PROPOSAL

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### AI-Powered Student Financial Document Management Platform: A Secure, Cloud-Native Approach

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| Module     | Research Design and Methodology (ITDMA3-22) |
| Faculty    | Information Technology                      |
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## SECTION A: INTRODUCTION

### 1. Research Title

#### **AI-Powered Student Financial Document Management Platform: A Secure, Cloud-Native Approach**

### 2. Background and Context

The volume of financial documentation that students are required to manage during their studies has grown considerably. Bursary agreements, NSFAS award letters, tuition fee statements, payment confirmations, appeal correspondence and bank letters constitute a significant body of personal financial records, many of which carry legal, financial or academic consequences. Despite the importance of these documents, students rarely have access to a structured system for managing them. In most cases, documents are stored informally across email accounts, messaging applications and personal devices, making retrieval difficult and increasing the risk that important deadlines or conditions go unnoticed.

Developments in Artificial Intelligence (AI) and Natural Language Processing (NLP) have substantially expanded the capabilities of document management systems. Traditional systems operated primarily as digital repositories, supporting file storage, folder organisation and keyword-based search. Contemporary AI systems can perform document classification, information extraction, summarisation and semantic question answering, offering a qualitatively different relationship between users and their documents. These capabilities are particularly relevant to student financial documents, which often contain complex conditions, varying terminology and time-sensitive obligations that students may not be equipped to interpret without assistance.

Retrieval-Augmented Generation (RAG) is an especially significant development in this context. Rather than relying on the general knowledge embedded in a language model, RAG retrieves relevant passages from a user's own document collection and uses those passages as context when generating a response. This means that when a student poses a question about their financial documents, the system can locate and present the actual answer from the relevant source, rather than producing a generalised or potentially inaccurate reply.

Cloud computing infrastructure enables systems of this complexity to be deployed at scale without prohibitive cost. Services such as Amazon Web Services (AWS) provide secure, scalable storage and compute resources that are well suited to document-intensive applications. A cross-platform approach, delivering both a web application and a mobile application, extends the practical value of such a system, given that many students primarily access digital services through mobile devices.

This research is positioned within the field of Software Engineering. It investigates how AI, cloud infrastructure, mobile development and security architecture can be combined into a cohesive, student-facing platform for financial document management. The research is both technical and applied in its orientation.

### **3. Problem Statement**

Students who receive financial assistance through NSFAS, institutional bursaries, private scholarships or student loans are required to engage with a substantial number of financial documents over the course of their studies. These documents often contain specific conditions, renewal requirements, appeal procedures and payment timelines. The consequences of missing or misunderstanding this information can include loss of funding, academic exclusion or unresolved debt.

Existing tools do not adequately address this problem. General-purpose cloud storage applications store files but do not interpret or analyse their content. University student portals provide limited document access and are not consistently optimised for mobile use. No dedicated platform currently exists that combines secure document storage, AI-assisted analysis and a mobile-first interface designed specifically around the financial document needs of students.

The problem also has a technical dimension. A platform that processes student financial documents must be designed with security as a foundational concern rather than an afterthought. Student financial data is highly sensitive and its misuse or exposure would cause real harm. Authentication, encryption, access control and audit logging must therefore be embedded into the system architecture from the outset, in alignment with the requirements of the Protection of Personal Information Act (POPIA).

Although the literature on AI document management is well developed in enterprise contexts, there is limited research specifically examining the application of these technologies to student financial document management. This gap both motivates and justifies the present study.

### **4. Research Questions**

#### **Main Research Question**

How can a cloud-native AI-powered platform be designed and implemented to support students in securely managing, understanding and acting on their financial documents?

#### **Supporting Research Questions**

- How can AI and NLP techniques be implemented to classify, summarise and extract information from student financial documents?
- How can Retrieval-Augmented Generation be applied to enable accurate, source-grounded question answering over a student's personal document collection?
- Which cloud-native architecture components are required to support secure document upload, storage and retrieval for a student user base?
- How should a web application and mobile application interact with backend AI services through a secure API layer?
- What authentication, encryption and audit controls are necessary to protect student financial data in compliance with POPIA?
- Which software engineering methods can be applied to evaluate the usability, accuracy, security and scalability of the proposed platform?

## **5. Research Objectives**

### **Main Objective**

To investigate the design and implementation of a cloud-native AI-powered document management platform that assists students in organising, understanding and securely managing their financial documents.

### **Specific Objectives**

1. To review AI and NLP techniques applicable to the classification, summarisation and information extraction of student financial documents.
2. To examine how Retrieval-Augmented Generation and vector search can support accurate question answering over a personalised student document repository.
3. To design a cloud-native system architecture that integrates web, mobile, backend, AI processing, storage and security components.
4. To evaluate technologies including Flutter, Next.js, Spring Boot, PostgreSQL and AWS S3 for suitability within the proposed system.
5. To define a document processing pipeline encompassing upload, OCR, chunking, embedding generation, indexing and retrieval.
6. To establish privacy and security controls that protect student financial data and align with POPIA compliance requirements.
7. To develop a research methodology that supports the evaluation of the proposed platform from technical, usability and adoption perspectives.

## **6. Justification for the Study**

This study is significant for several interconnected reasons. From a technical perspective, it requires the integration of multiple software engineering disciplines: cloud computing, mobile application development, backend API design, AI integration, database architecture and security engineering. The proposed platform therefore reflects the kind of

multidisciplinary system design that characterises contemporary enterprise software projects and provides a substantive basis for academic research at NQF Level 6.

From a social perspective, the study addresses a genuine and consequential problem. Students who rely on financial aid are among the most vulnerable members of a higher education institution. The inability to locate, understand or act on critical financial documents can have lasting academic and financial consequences. A platform that meaningfully reduces this burden is of both technical and social value.

The South African regulatory context further strengthens the justification. POPIA imposes specific legal obligations on any system that processes personal information, including the financial records of students. Research that examines how POPIA-compliant architecture can be embedded into an AI document platform addresses a gap in both the academic literature and in South African software development practice.

From an academic perspective, the study contributes a structured implementation framework for AI document intelligence in a student context, a contribution that extends existing enterprise-focused literature into a new and underexplored domain.

## SECTION B: LITERATURE REVIEW

### 7. Introduction to the Literature

The literature relevant to this study spans six intersecting technical domains: cloud-native architecture, artificial intelligence and natural language processing, retrieval-augmented generation, web and mobile application development, enterprise security and privacy compliance, and identified implementation gaps. The review does not seek to establish merely that AI is beneficial in document management in the abstract. Its purpose is to identify how specific technologies and design patterns can be implemented as components of a secure, student-facing financial document platform.

Each thematic section below informs one or more layers of the proposed system architecture. The review concludes by identifying the specific gaps in existing literature that this study is designed to address.

#### 7.1 Cloud-Native Enterprise Architecture

Cloud-native architecture describes the practice of designing systems specifically to operate within cloud infrastructure, taking advantage of elastic compute, managed storage, automated deployment and distributed processing. For a student-facing document platform, this approach is appropriate because usage patterns are irregular. Demand may increase substantially during bursary application cycles, examination periods or institutional payment deadlines, and a cloud-native design can scale to accommodate this without over-provisioning permanent infrastructure.

The AWS Well-Architected Framework provides a widely applied reference for evaluating cloud system design across six pillars: operational excellence, security, reliability, performance efficiency, cost optimisation and sustainability (Amazon Web Services, 2026). For the proposed platform, the security and reliability pillars are of particular importance given the sensitivity of student financial data. Object storage services such as Amazon S3 are well suited to document repositories, as they offer durable and scalable storage with server-side encryption enabled by default and support for customer-managed key encryption in high-sensitivity configurations (Amazon Web Services, 2026).

Containerisation using Docker and modular service design are also relevant architectural considerations. Separating the backend into discrete modules, such as authentication, document ingestion, AI processing and audit logging, improves maintainability and allows individual components to scale independently. For an academic prototype, a modular monolith using Spring Boot represents a practical approach that reduces deployment complexity while preserving clear separation of concerns. Production deployment could later decompose high-volume services such as embedding generation into independent microservices.

## 7.2 Artificial Intelligence and NLP for Document Intelligence

Artificial Intelligence and Natural Language Processing constitute the analytical layer of the proposed system. In the context of student financial documents, NLP capabilities support document classification, the extraction of named entities such as amounts, dates, conditions and parties, the generation of plain-language summaries and the detection of items requiring the student's attention.

Transformer-based language models, including BERT and its domain-adapted derivatives, represent the current state of the art for contextual text understanding. Models such as FinBERT, adapted for financial language, are particularly relevant for documents involving monetary values, interest rates and repayment obligations (Kim and Yang, 2024). Large Language Models extend these capabilities by enabling natural-language generation, allowing the system to produce coherent summaries and responses rather than only classifying or tagging content.

A key implementation principle is that AI functionality should be managed through the backend service layer rather than being invoked directly from client applications. This allows the platform to validate permissions, log AI interactions, control model parameters and prevent direct client access to sensitive document content. Human-in-the-loop design is equally important: AI-generated outputs should be presented as informational aids with source references, rather than as authoritative decisions. This is particularly relevant for student financial documents where misinterpretation could carry academic or financial consequences.

## 7.3 Retrieval-Augmented Generation and Vector Search

Retrieval-Augmented Generation is among the most relevant design patterns for an AI document platform in a student context. The RAG architecture connects a language model to a document repository so that responses are grounded in content the user has actually uploaded, rather than in the model's general training data. This reduces the risk of hallucination and enables the system to cite the specific document passages from which an answer is derived.

The implementation pipeline typically proceeds as follows: documents are uploaded and stored in object storage; text is extracted using parsers or OCR; extracted text is divided into chunks; each chunk is converted into a dense vector representation (embedding); the embeddings are stored in a vector database; and when a user submits a query, the query is similarly embedded and compared against stored vectors to retrieve the most semantically relevant passages. Those passages are then supplied to the language model as context for answer generation.

Vector databases such as Pinecone, Weaviate and ChromaDB support efficient similarity search at scale. For a prototype, an open-source vector store may be sufficient. For production deployment, a managed service reduces operational overhead while providing improved reliability and access control (Prasad, Boughanem and Dkaki, 2024). Access control during retrieval is a critical design requirement: the system must enforce document-level permissions so that a student's query can only retrieve content from documents they are authorised to view. This is both a security and a privacy obligation.

## **7.4 Web and Mobile Application Development**

The proposed platform requires both a web application and a mobile application. Students interact with financial documents in different contexts and on different devices. A web interface supports document management tasks that benefit from a larger screen, such as reviewing detailed summaries, managing account settings or uploading multiple files simultaneously. A mobile interface supports time-sensitive or location-independent tasks such as document capture, push notifications about upcoming deadlines and quick AI-assisted summaries.

Flutter is appropriate for the mobile application because it supports cross-platform development for both Android and iOS from a single codebase, using a consistent UI framework and a compiled rendering engine (Google, 2026). This reduces development effort and ensures a consistent user experience. Next.js is appropriate for the web application because it provides a full-stack React framework with built-in routing, server-side rendering and performance optimisation features (Vercel, 2026).

A fundamental architectural principle is that neither client application should communicate directly with cloud storage or AI services. Both should authenticate through an identity provider and interact with the platform exclusively through the backend API. The backend is then responsible for validating permissions, generating secure upload URLs, managing document metadata and orchestrating AI interactions. This reduces the attack surface and ensures that all significant actions are subject to authorisation and audit controls.

## **7.5 Enterprise Security, Privacy and Compliance**

Security and privacy represent the most critical non-functional requirements of the proposed platform. Student financial documents contain sensitive personal information including financial account details, funding conditions and correspondence with financial institutions. A privacy-by-design approach is therefore appropriate, meaning that security controls are embedded into the system architecture from the outset rather than retrofitted at a later stage.

Authentication and authorisation should be implemented using established standards. OAuth 2.0 provides a widely adopted framework for delegated authorisation, enabling



applications to obtain access tokens with limited, defined scope (IETF, 2012). OpenID Connect extends OAuth 2.0 with an identity layer, enabling client applications to verify user identity and obtain profile information through a standardised flow (OpenID Foundation, 2023). In practice, an identity provider such as Keycloak or AWS Cognito can manage token issuance, role assignment and multi-factor authentication.

Role-Based Access Control (RBAC) defines system-level permissions, while document-level metadata filters enforce per-student access boundaries. Encryption must be applied to data both at rest and in transit. Raw documents in object storage should be encrypted using server-side encryption, with customer-managed keys considered for high-sensitivity deployments. All API traffic must use HTTPS/TLS. Secrets such as API keys and database credentials must be stored in a secrets manager and must not appear in application code.

POPIA compliance imposes specific obligations with direct relevance to the proposed platform. These include processing personal information only for a defined and lawful purpose, minimising the collection of data to what is necessary, implementing appropriate technical safeguards, maintaining records of processing activities, and providing mechanisms for data subjects to access, correct or delete their information. The proposed system addresses each of these obligations through data minimisation, audit logging, retention controls and clearly defined deletion workflows.

AI-specific security considerations are also relevant. In RAG systems, prompt injection, unauthorised retrieval and data leakage represent known risk vectors. The platform must filter retrieved content by user permissions before it is included in any prompt, sanitise user inputs, separate system instructions from user-generated content, and maintain logs of all AI interactions for audit purposes (Preciado Martinez et al., 2025).

## **7.6 Identified Gaps in Existing Literature**

The existing literature establishes clearly that AI can improve document classification, search and compliance workflows in enterprise environments. However, there are several gaps that the present study is positioned to address.

First, the majority of research on AI document management is oriented toward enterprise users in sectors such as law, finance, insurance and government. There is limited research on the application of these technologies to student financial document management, which constitutes a distinct use case with different document types, user characteristics and risk profiles.

Second, mobile-first design is underrepresented in the AI document management literature. Most systems discussed assume a desktop or web-based interface. Given that many students access digital services primarily through mobile devices, this gap is significant for the proposed research.

Third, the South African regulatory context has received limited attention in the international literature. POPIA creates specific legal obligations that differ in some respects from GDPR, particularly in their application to South African institutions. Research examining how POPIA-compliant architecture can be embedded into AI document platforms is currently limited.

Fourth, the literature often evaluates AI document capabilities in isolation, without addressing the full implementation architecture. Studies may evaluate a classification model's accuracy without examining how it integrates with storage, authentication, mobile access and audit systems. The present study addresses this by proposing an integrated architecture and evaluation framework.

## **7.7 Theoretical and Technical Framework**

The study is supported by three interconnected frameworks. The Technology Acceptance Model (TAM) is relevant because adoption of the proposed platform depends on students perceiving it as useful and sufficiently easy to use (Davis, 1989; Venkatesh and Davis, 2000). These constructs will inform the design of the user-facing components and the evaluation criteria applied to usability.

Design Science Research provides the methodological rationale for producing a software artefact as a research output (Hevner et al., 2004). The study does not merely describe an existing problem; it proposes, designs and evaluates a practical solution. The technical framework is grounded in cloud-native architecture, RAG, secure API design, mobile-first development and privacy-by-design principles. These concepts collectively shape the proposed system and inform the evaluation criteria used in the methodology.

## SECTION C: PROPOSED SYSTEM ARCHITECTURE

### 8. System Overview

The proposed platform is designed as a layered cloud-native system in which each layer has a defined responsibility and communicates with adjacent layers through controlled interfaces. This architectural approach supports separation of concerns, independent scaling of components and clearer security boundaries. The system delivers functionality through a web application and a mobile application, both of which interact exclusively with the backend through a secure API.

| Layer                | Main Technologies                                  | Purpose                                                                                                                                         |
|----------------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Client Layer         | Next.js web app;<br>Flutter mobile app             | Delivers student-facing interfaces for document upload, AI-assisted search, financial document summaries, notifications and account management. |
| API Layer            | Spring Boot REST API; JWT validation               | Manages business logic, permission enforcement and communication between client applications and backend services.                              |
| Identity Layer       | Keycloak or AWS Cognito; OAuth 2.0; OpenID Connect | Handles student authentication, token issuance, role assignment, MFA and session management.                                                    |
| AI Processing Layer  | LLM API; LangChain or custom orchestration         | Generates document summaries, classifications, extracted financial details, risk indicators and question answers.                               |
| Retrieval Layer      | Embeddings; vector database; metadata filters      | Supports semantic search and RAG over each student's authorised document collection.                                                            |
| Storage Layer        | AWS S3; PostgreSQL; audit log tables               | Stores raw financial documents, structured metadata, user records and immutable event history.                                                  |
| Infrastructure Layer | Docker; CI/CD pipeline; AWS deployment; monitoring | Supports consistent deployment, horizontal scaling, observability and operational management.                                                   |

### 9. Proposed Technology Stack

The technology stack has been selected to balance implementation feasibility within a student project context with demonstrable relevance to industry practice. Each component

has a defined role within the architecture and can, where necessary, be substituted with an equivalent technology without restructuring the overall design.

| Component           | Recommended Choice                  | Reason for Selection                                                                                            |
|---------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Web Application     | Next.js                             | Provides a React-based framework with server-side rendering, routing and full-stack development capabilities.   |
| Mobile Application  | Flutter                             | Supports Android and iOS development from a single codebase with a consistent compiled UI.                      |
| Backend API         | Spring Boot                         | Well suited to enterprise REST API development with strong support for modular design and security integration. |
| Relational Database | PostgreSQL                          | Stores student metadata, document records, permissions, workflow states and audit logs reliably.                |
| Object Storage      | AWS S3                              | Provides scalable, durable and encrypted storage for raw financial documents and processed artefacts.           |
| Vector Database     | Pinecone, Weaviate or ChromaDB      | Stores document embeddings and supports efficient semantic retrieval for RAG.                                   |
| AI Orchestration    | LangChain or custom service layer   | Coordinates chunk retrieval, prompt construction and LLM API interactions.                                      |
| Authentication      | Keycloak or AWS Cognito             | Implements OAuth 2.0, OpenID Connect, role management, token validation and MFA.                                |
| Deployment          | Docker with AWS ECS/EKS or EC2      | Enables repeatable, containerised deployment with capacity for future service decomposition.                    |
| Monitoring          | AWS CloudWatch and application logs | Provides observability, incident detection and performance tracking.                                            |

## 10. Implementation Workflow

### 10.1 Document Upload and Storage

The upload workflow is initiated when a student selects a document through the web or mobile application. The client application does not communicate directly with cloud storage. Instead, it requests an upload session from the backend API. The backend verifies the student's identity and permissions, creates a document metadata record in PostgreSQL and generates a time-limited, pre-signed upload URL for the target S3 bucket. The client

then uploads the file directly to S3 using that URL. This design prevents unauthorised storage access and ensures that the backend maintains control over all document classification and metadata creation.

Following upload, the system records document attributes including name, type, owner identifier, upload timestamp, sensitivity classification and processing status. The raw file resides in S3 while searchable metadata is maintained in PostgreSQL.

## **10.2 Text Extraction and OCR**

Once a document has been stored, it enters the processing pipeline. Text-native formats such as searchable PDFs and DOCX files can be parsed directly to extract machine-readable content. Scanned documents and photographic images require Optical Character Recognition (OCR) to convert visual text into a processable form. The quality of this extraction step is significant because all downstream AI analysis depends on the accuracy of the extracted text. The system records extraction quality indicators, such as OCR confidence scores and parsing warnings, so that students can be informed if AI outputs for a particular document may be less reliable.

## **10.3 Chunking and Embedding Generation**

Following text extraction, documents are divided into smaller segments referred to as chunks. Chunking is necessary because language models operate within a defined context window and because semantic retrieval is more accurate when applied to focused passages than to entire documents. Each chunk is tagged with metadata including the source document identifier, page reference, section heading and sensitivity level, enabling filtered retrieval at query time.

Each chunk is then converted into a dense vector embedding using an embedding model. These embeddings are stored in the vector database alongside their associated metadata. When a student submits a search query or question, the query is embedded using the same model and compared against stored chunk embeddings to identify the most semantically relevant passages.

## **10.4 AI Analysis and RAG Question Answering**

The AI processing layer provides several functions relevant to student financial documents. Summarisation produces concise, plain-language overviews of document content, including key conditions, obligations and deadlines. Classification identifies the document type, such as a bursary agreement, NSFAS award letter, fee statement or payment confirmation. Entity extraction identifies and structures named elements including monetary amounts, payment dates, parties named and specific conditions or requirements.

Question answering employs the RAG pipeline. When a student submits a question, the backend validates the student's access rights, retrieves the most relevant chunks from their document collection using vector similarity search, constructs a prompt that includes the retrieved passages as context, and submits the prompt to the language model. The response returned to the student includes citations identifying the specific document sections from which the answer was derived, enabling independent verification.

10.5 Workflow Automation

Beyond document storage and AI analysis, the platform supports configurable workflow automation. Where a document contains a deadline, the system may generate a notification or calendar reminder. Where a document is identified as requiring a response or signature, the system may create an action item. These workflows are designed to support the student's own decision-making rather than to substitute for it. Final action on any document remains with the student, and AI-generated suggestions are presented with clear source references and confidence indicators.

11. Security Architecture

The security architecture is designed around five principles: verified identity, strict access control, comprehensive encryption, complete audit coverage and safe AI design. These principles are not implemented as optional features; they are embedded in the system architecture as foundational requirements.

| Security Control      | Implementation Approach                                  | Reason                                                                                                                           |
|-----------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Authentication        | OAuth 2.0 and OpenID Connect via Keycloak or AWS Cognito | Provides standardised, token-based identity verification for all student interactions with the platform.                         |
| Authorisation         | RBAC with student-level document filtering               | Ensures that each student can access only their own financial documents and that cross-student access is structurally prevented. |
| Encryption at Rest    | S3 server-side encryption; encrypted database storage    | Protects stored financial documents and metadata against unauthorised access to the underlying storage layer.                    |
| Encryption in Transit | HTTPS/TLS for all client, API and service communication  | Protects data during transmission across networks.                                                                               |
| Audit Logging         | Immutable event logs for all user and AI actions         | Supports accountability, POPIA compliance, breach investigation and regulatory reporting.                                        |

| Security Control   | Implementation Approach                                                                    | Reason                                                                                                                 |
|--------------------|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| AI Safety Controls | Permission-filtered retrieval; prompt logging; source citations                            | Prevents the AI layer from retrieving or exposing documents outside a student's authorised scope.                      |
| Human Oversight    | AI outputs presented as suggestions with source references                                 | Preserves student agency and prevents the automation of decisions with significant financial or academic consequences. |
| Data Minimisation  | Collect only data necessary for defined functions; enforce retention and deletion policies | Aligns with POPIA requirements and reduces harm in the event of a security incident.                                   |

## **SECTION D: RESEARCH METHODOLOGY**

### **12. Research Paradigm**

This study is guided by a pragmatist research paradigm, supported by a Design Science Research approach. Pragmatism is appropriate because the primary concern of the study is the design and evaluation of a practical solution to an identified problem, rather than the validation of a theoretical proposition. The value of the research is therefore measured in terms of the usefulness, feasibility and quality of the proposed artefact.

Design Science Research is appropriate because the central output of the study is an artefact: a technical architecture and implementation framework for a student financial document management platform. The research combines theoretical analysis, requirements elicitation and practical system design, and the artefact is evaluated against defined criteria (Hevner et al., 2004). This positions the study within the tradition of applied software engineering research.

### **13. Research Method**

The study employs a mixed-methods design, combining quantitative data collection, qualitative inquiry and a technical design component. The quantitative phase identifies the most common student document management challenges and the AI features of greatest perceived value. The qualitative phase examines student expectations, trust concerns and workflow requirements in greater depth. The technical component involves the design and evaluation of the proposed system architecture against defined software engineering criteria.

This combination of methods allows the study to address both the human and technical dimensions of the research problem. The methodology is consistent with an applied software engineering study at NQF Level 6, where technical design must be informed by and evaluated against real user requirements.

### **14. Data Collection Methods**

#### **14.1 Technical Literature and Documentation Review**

The first phase of data collection involves a systematic review of academic literature, industry white papers and official technology documentation relevant to AI document processing, RAG, cloud-native architecture, mobile development, authentication standards, database design and security frameworks. This review serves two functions: it provides the theoretical and technical foundation for the proposed architecture, and it identifies the specific implementation gaps that the study addresses.



## **14.2 Quantitative Survey**

An online survey will be administered to students who manage financial aid documentation as part of their studies. The target sample comprises 100 to 150 respondents drawn from students receiving NSFAS funding, institutional bursaries, private scholarships or student loans. The survey will measure current document management practices, frequency of document-related difficulties, the perceived usefulness of specific AI features, preferred access platforms and concerns relating to data privacy and security.

Likert-scale items will be used to enable statistical analysis. The survey instrument will be reviewed by the research team prior to distribution to ensure clarity and alignment with the research questions.

## **14.3 Qualitative Interviews**

Following the survey, 12 to 18 participants will be selected for semi-structured interviews. Participants will be chosen purposively to represent a range of financial aid types and levels of technical familiarity. The interviews will explore topics that the survey instrument cannot adequately capture, including students' experiences with specific document management challenges, their attitudes toward AI-assisted tools, their expectations regarding data privacy and their preferred workflows for document review and action.

Interviews will be conducted online using Microsoft Teams or Zoom, recorded with participant consent and transcribed for analysis. A semi-structured format will be used to ensure consistency across interviews while allowing participants to raise issues not anticipated in the interview guide.

## **14.4 Prototype and Architecture Evaluation**

Where feasible within the scope of the project, a low-fidelity or partial prototype will be developed to demonstrate core platform workflows, including document upload, text extraction, AI summarisation, semantic search and authentication. Where a complete functional prototype is not achievable, the study will produce architecture diagrams, data flow models and implementation specifications that serve as the primary technical artefact.

The architecture will be evaluated against a structured checklist informed by cloud-native design principles, security requirements, POPIA compliance and usability considerations. This ensures that the proposed system is assessed as a software engineering artefact rather than as a theoretical concept.

# **15. Data Analysis Methods**

## **15.1 Quantitative Analysis**

Survey data will be analysed using descriptive statistics, including frequency distributions, percentages, means and standard deviations. These measures will identify the most prevalent student document management challenges and the most valued AI capabilities. Where appropriate, inferential tests will be applied to examine relationships between variables. Chi-square tests may be used to determine whether document management challenges vary by financial aid type. Correlation analysis may examine relationships between perceived usefulness, perceived ease of use and willingness to adopt an AI document tool. Statistical analysis will be conducted using SPSS or an equivalent package.

### **15.2 Qualitative Analysis**

Interview transcripts will be analysed using the thematic analysis approach described by Braun and Clarke (2006). Initial codes will be generated from participant responses and subsequently grouped into themes. Analysis will be both deductive, drawing on constructs from TAM, security frameworks and cloud architecture literature, and inductive, allowing emergent themes to develop from participant data. This approach ensures that the analysis remains theoretically grounded while remaining responsive to the full range of student experience.

### **15.3 Technical Evaluation**

The proposed architecture will be evaluated against defined technical criteria, including functional completeness, security coverage, scalability, maintainability, platform accessibility and AI retrieval quality. If a prototype is developed, performance indicators such as document upload processing time, semantic search response time and AI summarisation latency can be recorded. AI output quality will be assessed through structured human review, examining whether summaries are accurate, whether retrieved passages are relevant and whether answers are appropriately grounded in source documents.

### **15.4 Integration of Findings**

Findings from the literature review, quantitative survey, qualitative interviews and technical evaluation will be integrated into a final implementation framework. This framework will specify recommended system components, student requirements, security controls, workflow patterns and evaluation criteria. Where findings converge across methods, recommendations will be strengthened accordingly. Where findings diverge, the discrepancies will be reported and their implications for the design discussed.

## **16. Handling Incomplete or Inconsistent Data**

Incomplete survey responses will be reviewed prior to analysis. Responses in which the majority of items are unanswered will be excluded. For isolated missing values in Likert-scale items, mean imputation may be applied where missingness does not appear to be systematic. Outlier responses will be identified through descriptive statistics and reviewed

individually; genuine extreme values will be retained, as they may reflect important user experiences.

In qualitative analysis, inconsistent statements within interview transcripts will not be treated as errors. They will be interpreted as evidence of complexity in participants' experiences. Where clarification is required, follow-up questions will be posed during the interview itself. Transcripts will be reviewed for completeness following each session.

## SECTION E: ETHICAL CONSIDERATIONS

### 17. Ethical Framework

This study involves human participants and engages with sensitive personal financial data. Ethical care is therefore central to the research design rather than a peripheral concern. The study will adhere to Eduvos research ethics requirements and will be guided by the principles of voluntary participation, informed consent, confidentiality, data protection, non-maleficence and research integrity, as well as the obligations imposed by POPIA on the processing of personal information.

All participants will be provided with a clear description of the study's purpose, the nature of the questions to be asked, the expected time commitment, the voluntary nature of participation and their right to withdraw at any stage without consequence. Survey participation will be anonymised where possible. Interview participants will provide explicit consent, whether written or verbal, before any recording commences.

#### 17.1 Core Ethical Commitment: Student Financial Data Privacy

As a group, we have identified the privacy and security of student financial data as the central ethical principle underpinning our application design. This commitment reflects our understanding that student financial documents, including bursary agreements, NSFAS correspondence, bank records and fee statements, contain highly sensitive personal information. The misuse, exposure or unauthorised access of this data could cause significant harm, including identity fraud, financial loss and reputational damage.

In practical terms, this commitment is expressed through several architectural decisions. Data minimisation governs the collection of personal information: the system collects only what is necessary to fulfil its defined functions, reducing the volume of sensitive data at risk in the event of a security incident. All documents stored in the system are encrypted at rest using server-side encryption, and all data transmitted between the application and the backend is protected using HTTPS/TLS. Access control is implemented at the individual student level, ensuring that cross-student document access is structurally prevented rather than merely restricted by policy.

Every significant user and AI action is recorded in an audit log. This includes document uploads, views, AI analysis requests, search queries, modifications and deletions. These logs are immutable and are retained in accordance with institutional policy. They provide a foundation for accountability and are essential for any investigation of a suspected breach or misuse.

AI interactions with student documents are subject to the same privacy principles. The AI layer may only retrieve and process documents that the authenticated student is authorised to access. AI-generated outputs are presented to the student as informational aids with source citations, and are not shared with third parties without explicit consent. The system does not repurpose student document data for model training or any function beyond the stated purpose of the platform.

POPIA compliance is treated as a legally enforceable and morally substantive requirement rather than a procedural formality. The platform design embeds the relevant obligations through purpose limitation, consent management where applicable, data subject access mechanisms and structured deletion workflows. These controls ensure that the platform's handling of student financial data is both technically sound and legally responsible.

## **17.2 Confidentiality and Research Risk**

The principal risk associated with the research process is that participants may disclose information about their own financial circumstances or about weaknesses in their institution's document management systems. To mitigate this risk, the study will avoid collecting identifying information beyond what is necessary, and findings will be reported at an aggregated or thematic level. Interview participants will be assigned pseudonyms in all research outputs. Organisation names, client identifiers and operationally sensitive details will be excluded from the final report.

Participants will be reminded at the outset of each interview that they should not disclose proprietary, confidential or legally restricted information. In the event that sensitive information is inadvertently disclosed, it will be excluded from the analysis and will not appear in any research output.

## **17.3 Data Protection During Research**

All research data, including recordings, transcripts and survey responses, will be stored in password-protected locations accessible only to the research team. Data will be retained for the minimum period required by Euvos institutional policy and will be securely deleted thereafter.

Any prototype demonstration will use synthetic or anonymised sample documents rather than real student financial records. No actual student data will be processed through external AI systems unless explicit, informed consent has been obtained and appropriate safeguards are in place. This requirement applies to all stages of development and testing.

## **17.4 AI Ethics**

The AI components of the proposed platform raise specific ethical considerations. AI-generated summaries, classifications and answers must be presented as assistive outputs rather than authoritative determinations. Students should be able to inspect the source passages underlying any AI response and should be expected to exercise independent judgement before acting on AI-generated information, particularly where financial or academic consequences are involved.

The research team is aware of the risks of hallucination, where a language model generates plausible but incorrect information, and of systematic bias, where model performance may vary across document types, languages or writing styles. These risks are addressed in the system design through permission-filtered retrieval, mandatory source citations, confidence indicators and comprehensive audit logging. The final report will document these risks transparently and discuss their implications for responsible deployment.

## **18. Research Integrity**

The research team is committed to the principles of honest and accurate research practice. Findings will be reported as they arise, including results that do not support the initial hypotheses or design expectations. Data will not be fabricated, selectively excluded or misrepresented. All sources consulted will be properly cited in accordance with the referencing conventions required by Eduvos.

A record of significant methodological decisions will be maintained throughout the study, including decisions relating to survey design, participant selection, qualitative coding and technical evaluation criteria. This documentation creates an audit trail of the research process and supports the credibility and transferability of the findings.

Any use of AI tools during the drafting, analysis or prototype development phases of the project will be disclosed in accordance with Eduvos requirements. The submitted work represents the research team's own analysis, design decisions and conclusions.

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